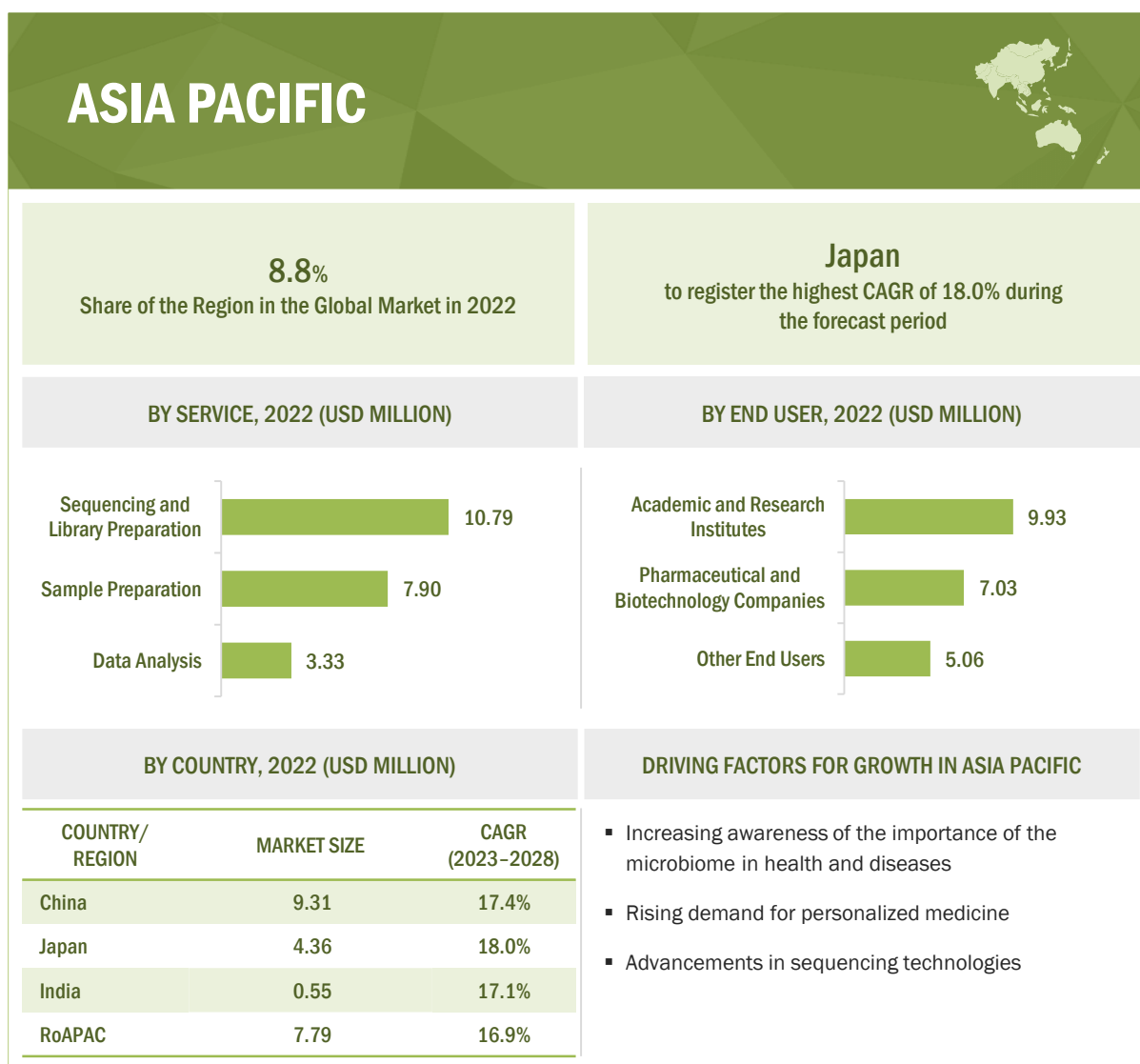


10.4 ASIA PACIFIC

The Asia Pacific microbiome sequencing services market is estimated to reach USD 57.19 million by 2028 from USD 25.73 million in 2023, at a CAGR of 17.3% during the forecast period. In 2022, China accounted for the largest share of 42.3% of the microbiome sequencing services market in the Asia Pacific, followed by Japan with a share of 19.8%.

The Asia Pacific region is expected to be the fastest-growing market for microbiome sequencing services due to the increasing awareness of the importance of the microbiome in health and disease, rising demand for personalized medicine, and advancements in sequencing technologies. Another key driver for the Asia Pacific microbiome sequencing services market is the region's large and rapidly aging population. As people age, they become more susceptible to a range of diseases and conditions that may be influenced by the microbiome. This has led to an increased focus on preventative healthcare and personalized medicine, which is driving the demand for microbiome sequencing services. The development of high-throughput sequencing technologies has made it possible to sequence the entire microbiome in a single sample, enabling researchers to more accurately and comprehensively characterize the microbial communities that are present in samples.

Challenges in terms of data protection and privacy, as well as the high cost of microbiome sequencing services, are expected to restrain the growth of the Asia Pacific market. In addition, there may be cultural and linguistic barriers to adoption in certain regions, as well as concerns about the accuracy and clinical utility of microbiome sequencing data.

FIGURE 27 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET SNAPSHOT

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 131 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	7.96	9.31	10.89	12.73	14.89	17.46	20.54	24.24	17.4%
Japan	3.70	4.36	5.12	6.02	7.08	8.34	9.86	11.70	18.0%
India	0.47	0.55	0.64	0.75	0.87	1.02	1.20	1.41	17.1%
Rest of Asia Pacific	6.68	7.79	9.08	10.57	12.32	14.39	16.86	19.83	16.9%
Total	18.81	22.01	25.73	30.07	35.16	41.22	48.46	57.19	17.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 132 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	6.76	7.90	9.22	10.76	12.56	14.71	17.27	20.35	17.2%
Sequencing and Library Preparation	9.17	10.79	12.68	14.91	17.53	20.67	24.44	29.01	18.0%
Data Analysis	2.89	3.33	3.83	4.40	5.06	5.84	6.75	7.83	15.4%
Total	18.81	22.01	25.73	30.07	35.16	41.22	48.46	57.19	17.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 133 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	7.79	9.16	10.75	12.63	14.83	17.47	20.63	24.46	17.9%
Whole-genome Sequencing	4.08	4.72	5.45	6.30	7.28	8.43	9.80	11.42	15.9%
Shotgun Sequencing	5.04	5.97	7.07	8.36	9.89	11.74	13.96	16.67	18.7%
Other Types	1.91	2.17	2.46	2.79	3.16	3.59	4.08	4.65	13.6%
Total	18.81	22.01	25.73	30.07	35.16	41.22	48.46	57.19	17.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 134 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	8.47	9.93	11.63	13.62	15.96	18.75	22.09	26.12	17.6%
Sequencing by Ligation	2.42	2.83	3.29	3.84	4.48	5.23	6.13	7.22	17.0%
Nanopore Sequencing	0.96	1.18	1.44	1.76	2.14	2.62	3.20	3.92	22.2%
Other Technologies	6.96	8.08	9.36	10.85	12.58	14.62	17.04	19.94	16.3%
Total	18.81	22.01	25.73	30.07	35.16	41.22	48.46	57.19	17.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 135 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	5.92	7.03	8.33	9.86	11.68	13.87	16.52	19.74	18.8%
Academic and Research Institutes	8.52	9.93	11.55	13.44	15.64	18.25	21.36	25.09	16.8%
Other End Users	4.37	5.06	5.85	6.77	7.84	9.10	10.59	12.36	16.1%
Total	18.81	22.01	25.73	30.07	35.16	41.22	48.46	57.19	17.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.4.1 CHINA

10.4.1.1 Increasing investments in R&D to drive growth

The microbiome sequencing services market in China has seen significant growth in recent years. The rapidly expanding healthcare industry in China has led to increased investment in the research and development of new microbiome sequencing technologies. China is home to one of the world's fastest-growing biotechnology markets. In 2020, Chinese genetic testing companies in the primary market raised over RMB 19.3 billion (USD 3 billion) in financing—an annual growth rate of 153% compared to 2019.

The rising incidence of chronic diseases in China, such as cancer and diabetes, has fueled the demand for more accurate and efficient diagnostic tools. Additionally, the Chinese government has placed a strong emphasis on promoting the development of the biotechnology and life science industries, which has created opportunities for companies in the microbiome sequencing services market to receive funding and support for R&D efforts. This has also encouraged collaborations between academic institutions and private companies in China, which has facilitated the development of new technologies and therapies. For instance, Biomica Ltd. announced the signing of a definitive agreement for a USD 20 million financing round to be led by Shanghai Healthcare Capital (SHC). The financing is subject to customary closing conditions, including clearance by Chinese regulatory authorities. The financing round will enable Biomica to forge ahead, developing its pipeline of microbiome-based therapeutics. Moreover, Siolta Therapeutics

announced that it raised USD 30 million in a series B investment round to support the clinical development of its lead product, STMC-103H. The series B funding involved participation from top-tier investors and venture firms, including series A investors Khosla Ventures and Marc Benioff, as well as new investors such as Seventure (Health for Life Capital Fund), SymBiosis, and Global Brain (Kirin Health Innovation Fund/GB-VII). Siolta's live biotherapeutic product (LBP) platform is designed to develop microbiome-based medicines and diagnostics for the prevention and treatment of chronic diseases in targeted populations.

One of the main restraints for the microbiome sequencing services market in China is the lack of regulatory oversight and standardization in the industry. This has created uncertainty for investors and may hinder the growth of the market in the long term. Additionally, the high cost of technology and limited reimbursement options may limit the accessibility of microbiome sequencing services among patients.

Despite these challenges, there still are several opportunities for the growth of the microbiome sequencing services market in China. For instance, the growing demand for personalized medicine and precision healthcare has increased the focus on individualized microbiome analysis, which has the potential to drive the demand for microbiome sequencing services. The rapid growth of the Chinese middle class has led to an increased demand for high-quality healthcare services, which could further accelerate the growth of the microbiome sequencing services market.

TABLE 136 CHINA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	2.86	3.34	3.90	4.56	5.32	6.23	7.32	8.63	17.2%
Sequencing and Library Preparation	3.88	4.56	5.36	6.30	7.41	8.74	10.33	12.26	18.0%
Data Analysis	1.22	1.41	1.62	1.87	2.15	2.49	2.89	3.36	15.7%
Total	7.96	9.31	10.89	12.73	14.89	17.46	20.54	24.24	17.4%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 137 CHINA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	3.30	3.88	4.55	5.34	6.27	7.39	8.72	10.34	17.8%
Whole-genome Sequencing	1.72	2.00	2.31	2.67	3.09	3.59	4.17	4.88	16.1%
Shotgun Sequencing	2.13	2.52	2.99	3.53	4.18	4.95	5.89	7.02	18.7%
Other Types	0.80	0.92	1.04	1.19	1.35	1.53	1.75	2.00	14.0%
Total	7.96	9.31	10.89	12.73	14.89	17.46	20.54	24.24	17.4%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 138 CHINA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	3.58	4.21	4.93	5.77	6.77	7.96	9.38	11.09	17.6%
Sequencing by Ligation	1.02	1.19	1.39	1.62	1.89	2.21	2.58	3.04	16.9%
Nanopore Sequencing	0.41	0.50	0.61	0.74	0.90	1.09	1.33	1.63	21.9%
Other Technologies	2.94	3.42	3.96	4.60	5.34	6.21	7.25	8.49	16.4%
Total	7.96	9.31	10.89	12.73	14.89	17.46	20.54	24.24	17.4%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 139 CHINA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	2.51	2.98	3.52	4.16	4.93	5.84	6.95	8.30	18.7%
Academic and Research Institutes	3.60	4.20	4.89	5.69	6.63	7.75	9.07	10.67	16.9%
Other End Users	1.84	2.14	2.48	2.87	3.33	3.87	4.51	5.28	16.3%
Total	7.96	9.31	10.89	12.73	14.89	17.46	20.54	24.24	17.4%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.4.2 JAPAN

10.4.2.1 Significant growth in geriatric population to present opportunities for market growth

The microbiome sequencing services market in Japan has been growing rapidly in recent years, driven by several factors. One key driver is the increasing prevalence of chronic diseases, such as cancer, diabetes, and IBD, which have been linked to dysbiosis or imbalances in the gut microbiome. This has led to a growing interest in the potential of microbiome analysis for personalized medicine and disease diagnosis and treatment. Another driver is the development of microbiome-based therapeutics, such as fecal microbiota transplantation (FMT), which have been approved in Japan for the treatment of recurrent *Clostridioides difficile* infections.

In addition to these drivers, Japan's aging population presents a unique opportunity for microbiome sequencing services. Japan's population is aging and shrinking fast. With a median age of 48.4 years, Japan's population is the oldest in the world. The government of Japan projects that there will be almost one elderly person for each person of working age by 2060. Over 29.18% of the total population is 65 years or older. As the population ages, the incidence of age-related diseases such as dementia and Parkinson's disease is increasing, and there is a growing interest in the potential of the gut microbiome to

play a role in these conditions. Japan's highly developed healthcare system and advanced research infrastructure also make it an attractive market for microbiome sequencing services.

However, there are several restraints to the growth of the microbiome sequencing services market in Japan. One is the lack of public awareness and understanding of the microbiome and its potential applications in healthcare. This has led to the slow adoption of microbiome analysis services by both patients and healthcare providers. In addition, regulatory hurdles and a lack of reimbursement for microbiome analysis services have hindered the growth of this market.

Despite these challenges, the increasing availability of NGS technologies and bioinformatics tools, which have greatly improved the speed and accuracy of microbiome analysis, are expected to offer potential growth opportunities for market growth in Japan.

TABLE 140 JAPAN: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	1.33	1.56	1.84	2.15	2.53	2.98	3.51	4.16	17.8%
Sequencing and Library Preparation	1.80	2.14	2.52	2.98	3.53	4.18	4.97	5.93	18.6%
Data Analysis	0.57	0.66	0.76	0.88	1.02	1.18	1.38	1.61	16.1%
Total	3.70	4.36	5.12	6.02	7.08	8.34	9.86	11.70	18.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 141 JAPAN: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	1.53	1.81	2.14	2.53	2.98	3.53	4.20	5.00	18.5%
Whole-genome Sequencing	0.80	0.93	1.09	1.26	1.47	1.71	2.00	2.34	16.6%
Shotgun Sequencing	0.99	1.18	1.41	1.67	1.99	2.37	2.84	3.41	19.3%
Other Types	0.37	0.43	0.49	0.56	0.64	0.73	0.83	0.95	14.3%
Total	3.70	4.36	5.12	6.02	7.08	8.34	9.86	11.70	18.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 142 JAPAN: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	1.67	1.97	2.32	2.73	3.21	3.80	4.50	5.34	18.2%
Sequencing by Ligation	0.48	0.56	0.66	0.77	0.90	1.06	1.25	1.47	17.6%
Nanopore Sequencing	0.19	0.23	0.29	0.35	0.43	0.53	0.65	0.80	22.8%
Other Technologies	1.37	1.60	1.86	2.17	2.53	2.96	3.47	4.08	17.0%
Total	3.70	4.36	5.12	6.02	7.08	8.34	9.86	11.70	18.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 143 JAPAN: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	1.17	1.39	1.66	1.97	2.35	2.80	3.36	4.03	19.5%
Academic and Research Institutes	1.68	1.97	2.30	2.69	3.15	3.70	4.35	5.14	17.4%
Other End Users	0.86	1.00	1.17	1.36	1.58	1.84	2.16	2.53	16.8%
Total	3.70	4.36	5.12	6.02	7.08	8.34	9.86	11.70	18.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.4.3 INDIA

10.4.3.1 Increasing prevalence of chronic diseases to support growth

The microbiome sequencing services market in India is experiencing significant growth due to several drivers. One of the major drivers is the increasing prevalence of chronic diseases in the country. According to the World Health Organization, in 2019, non-communicable diseases such as cardiovascular diseases, diabetes, and cancer accounted for 66% of all deaths in India. The microbiome has been shown to play a crucial role in the development and progression of these diseases, making microbiome sequencing services an important tool in disease prevention and management.

Another driver is the growing demand for personalized medicine. With advancements in microbiome research, there is a growing understanding of the unique microbiome profiles of individuals and how they can impact their health. Microbiome sequencing services can provide personalized insights into an individual's microbiome composition and help in developing personalized treatment plans.

Government initiatives and funding programs are also driving the growth of the microbiome sequencing services market in India. The Department of Biotechnology (DBT) has launched several programs to promote microbiome research in the country, such as the Human Microbiome Project and the Microbial

Culture Collection program. These initiatives have created opportunities for the development of new microbiome sequencing services and technologies in India.

One of the major restraints for market growth in India is the lack of awareness and understanding of the microbiome among the public and healthcare professionals. This is limiting the adoption of microbiome sequencing services and their incorporation into clinical practice.

TABLE 144 INDIA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	0.17	0.20	0.23	0.27	0.31	0.36	0.43	0.50	16.9%
Sequencing and Library Preparation	0.23	0.27	0.32	0.37	0.44	0.51	0.61	0.72	17.8%
Data Analysis	0.07	0.08	0.10	0.11	0.13	0.14	0.17	0.19	15.0%
Total	0.47	0.55	0.64	0.75	0.87	1.02	1.20	1.41	17.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 145 INDIA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	0.19	0.23	0.27	0.31	0.37	0.43	0.51	0.60	17.7%
Whole-genome Sequencing	0.10	0.12	0.14	0.16	0.18	0.21	0.24	0.28	15.6%
Shotgun Sequencing	0.13	0.15	0.18	0.21	0.25	0.29	0.35	0.41	18.5%
Other Types	0.05	0.05	0.06	0.07	0.08	0.09	0.10	0.11	13.2%
Total	0.47	0.55	0.64	0.75	0.87	1.02	1.20	1.41	17.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 146 INDIA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	0.21	0.25	0.29	0.34	0.40	0.46	0.55	0.64	17.3%
Sequencing by Ligation	0.06	0.07	0.08	0.10	0.11	0.13	0.15	0.18	16.8%
Nanopore Sequencing	0.02	0.03	0.04	0.04	0.05	0.07	0.08	0.10	22.2%
Other Technologies	0.17	0.20	0.23	0.27	0.31	0.36	0.42	0.49	16.0%
Total	0.47	0.55	0.64	0.75	0.87	1.02	1.20	1.41	17.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 147 INDIA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	0.15	0.18	0.21	0.25	0.29	0.35	0.41	0.49	18.7%
Academic and Research Institutes	0.21	0.25	0.29	0.33	0.39	0.45	0.53	0.62	16.5%
Other End Users	0.11	0.13	0.15	0.17	0.19	0.23	0.26	0.30	15.8%
Total	0.47	0.55	0.64	0.75	0.87	1.02	1.20	1.41	17.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.4.4 REST OF ASIA PACIFIC

The RoAPAC includes Australia, Taiwan, New Zealand, Thailand, South Korea, Singapore, and Malaysia, among other Asia Pacific countries. Factors such as public and private initiatives to support the adoption and development of microbiome technologies, increasing R&D expenditure, and increasing awareness of the diagnostic applications of microbiome sequencing are driving the growth of the RoAPAC market.

In September 2019, the South Korean government proposed plans to increase the budget of government subsidies in the biotech sector by 16% and finance a project to secure the genome data of 1 million people. The government aims to increase its budget spending to help the pharmaceutical sector develop innovative new drugs and medical devices. As per this agenda, the government will invest USD 951.8 million in R&D projects in the sector in 2020, with an increase of 16% in 2019. Over the past few years, a few events focusing on spreading awareness about the benefits of microbiome sequencing and the latest advancements in this field have been organized in the RoAPAC region.

In June 2022, Singapore-based precision gut microbiome company AMILI received USD 10.5 million in a Series A funding round led by technology and life sciences investor Vulcan Capital. New investors also joined the round, including Pruksha Group, TVM Capital Healthcare, Emtex Group, Capital Code, Pureland Group, Blue7, and Enterprise Singapore's investment arm SEEDS Capital.

TABLE 148 REST OF ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	2.40	2.79	3.25	3.78	4.40	5.13	6.00	7.05	16.8%
Sequencing and Library Preparation	3.25	3.82	4.48	5.25	6.16	7.24	8.54	10.10	17.7%
Data Analysis	1.03	1.18	1.35	1.54	1.76	2.02	2.32	2.68	14.7%
Total	6.68	7.79	9.08	10.57	12.32	14.39	16.86	19.83	16.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 149 REST OF ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	2.76	3.24	3.79	4.44	5.20	6.11	7.20	8.51	17.6%
Whole-genome Sequencing	1.45	1.67	1.92	2.21	2.54	2.93	3.38	3.92	15.3%
Shotgun Sequencing	1.79	2.12	2.50	2.95	3.48	4.12	4.89	5.82	18.5%
Other Types	0.68	0.77	0.87	0.97	1.10	1.23	1.39	1.57	12.7%
Total	6.68	7.79	9.08	10.57	12.32	14.39	16.86	19.83	16.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 150 REST OF ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	3.01	3.51	4.10	4.78	5.58	6.53	7.67	9.03	17.1%
Sequencing by Ligation	0.86	1.00	1.17	1.36	1.58	1.84	2.15	2.53	16.7%
Nanopore Sequencing	0.34	0.42	0.51	0.63	0.76	0.93	1.14	1.40	22.2%
Other Technologies	2.48	2.86	3.30	3.81	4.40	5.09	5.90	6.88	15.8%
Total	6.68	7.79	9.08	10.57	12.32	14.39	16.86	19.83	16.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 151 REST OF ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	2.10	2.49	2.94	3.48	4.11	4.88	5.80	6.92	18.7%
Academic and Research Institutes	3.03	3.52	4.07	4.72	5.47	6.36	7.41	8.66	16.3%
Other End Users	1.55	1.79	2.06	2.38	2.74	3.16	3.66	4.25	15.6%
Total	6.68	7.79	9.08	10.57	12.32	14.39	16.86	19.83	16.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.4.5 ASIA PACIFIC: IMPACT OF RECESSION

The impact of the recession on the global market would affect the Asia Pacific region as well. The Association of Southeast Asian Nations (ASEAN) projected a steady decline in the economic growth of Asia Pacific countries owing to the slower growth of their trading partners, namely, Europe, the US, and China. The overall GDP decline in the coming years is expected to affect the healthcare expenditure, spending, and wages in the industry with reduced import and export activities.

However, in Asia, countries focus on increasing their R&D in the pharmaceutical and biotechnology industry. India has seen a progression in terms of its R&D funding and rising government support. According to a report by India Brand Equity Foundation (IBEF), India's domestic pharmaceutical market was estimated at USD 42 billion in 2021. It is likely to reach USD 65 billion by 2024 and may further expand to reach USD 120–130 billion by 2030. Sun Pharma, a leading pharmaceutical company in India, announced that it intends to invest USD 600–650 million in R&D. On the other hand, Chinese pharmaceutical companies invested nearly 10% of their sales revenue in R&D in 2021.

For the overall Asia Pacific region, the impact of the recession is expected to be low during the forecast period. Japan's economy is expected to be more stable than other Asian countries. However, price inflation leading to lower consumption would lead to a slight decline in Japan's economic growth in 2023.

10.5 LATIN AMERICA

The Latin American microbiome sequencing services market is expected to reach USD 8.88 million by 2028 from USD 5.26 million in 2023, at a CAGR of 11.1% from 2023 to 2028.

The Latin American microbiome sequencing services market has been growing steadily in recent years. The region is witnessing an increased prevalence of chronic diseases, such as cancer, IBD, and diabetes, which have been linked to imbalances in the gut microbiome. As a result, there is a growing demand for microbiome sequencing services to better understand and manage these conditions. Additionally, the region's large and diverse population provides a unique opportunity for microbiome research and discovery.

The lack of awareness and understanding of the potential benefits of microbiome sequencing among healthcare providers and the general population is a key factor restraining the growth of this regional market. This has resulted in a relatively small market size and limited investment in the field. Additionally, the high cost of sequencing and the lack of standardization and regulation of microbiome testing can be a barrier to entry for smaller companies.